

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. (CL) Examination

December-2013

MA 703: Applications of Numerical Methods in Structural Engineering

Date: 30.12.2013, Monday

Time: 10:00 a.m. To 01:00 p.m.

Total: 70 Marks

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

Q-1

(4+3)

(a) Find Fourier Sine Transform of the function $f(t) = \frac{e^{-at}}{t}$

(b) Find Fourier Transform of the function $f(t) = e^{-k|t|}$, $k > 0$

Q-2

(6+8)

(a) Find solution of $x^2 - 5x - 6 = 0$ using Bisection method, correct up to two decimal places. Use $a = 5.9$ and $b = 6.05$ as initial approximations.

(b) Find out solution of $x^3 - 125 = 0$ using Regula - Falsi method, correct up to three decimal places. Use $a = 3$ and $b = 6$ as initial approximations.

OR

Q-2

(7+7)

(a) Find out solution of $x^4 - 64 = 0$ using Newton Raphson method, correct up to four decimal places. Use $a = 2$ as initial approximation.

(b) Use Newton's Forward Interpolation formula, find the value of $f(1.6)$

x	1	1.4	1.8	2.2
$f(x)$	3.49	4.82	5.96	6.5

Q-3

(6+6+2)

(a) Find out solution of $\frac{dy}{dx} = 1 + xy$ up to third approximation. Using Picard's Method given that $y(0) = 0$

- (b) Apply Lagrange's Interpolation formula to find $y(3)$ using points $(1,1)$, $(2,4)$, $(4,16)$.
- (c) Write down Newton's Divided difference interpolation formula.

OR

Q-3

(7+7)

- (a) Use Euler's method to find $y(1)$ from the differential equation: $\frac{dy}{dx} = x + y$, $y(0) = 1$
[take $h = 0.1$]
- (b) Using Runge-Kutta method of second order, find the approximate value of y at $x = 1.2$,
taking $h = 0.1$, where $\frac{dy}{dx} = x^2 + y^2$; $y(1) = 0$

SECTION-II

Q-4

(4+3)

- (a) Define: Analytic function and orthogonal system. Show that every analytic function $f(z) = u + iv$ defines two families of curves $u(x, y) = 0$ and $v(x, y) = 0$ which forms an orthogonal system.
- (b) Prove polar form of Cauchy-Riemann equations.

Q-5

(5+5+4)

- (a) Construct an analytic function which satisfied $u - v = e^x (\cos x - \sin x)$
- (b) Verify that $A = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$ is Unitary matrix.
- (c) Suppose A and B are two Symmetric matrices then prove that AB is Symmetric if and only if A and B are Commute.

OR

Q-5

(5+5+4)

- (a) Express $A = \begin{bmatrix} 2+6i & 5+3i \\ 9-i & 4-2i \end{bmatrix}$ as sum of Hermitian and Skew-Hermitian Matrices.
- (b) Prove that inverse of Unitary matrix is Unitary

(c) If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$ then prove that $A^2 - 4A - 5I$ is Null matrix.

Q-6

(4+3+7)

(a) Calculate the coefficient of correlation from the following data also find its probable error:

X	100	200	300	400	500	600	700
Y	30	50	60	80	100	110	130

(b) From the following table calculate the Karl-Pearson's coefficient of correlation:

X	6	2	10	4	8
Y	9	11	?	8	7

Arithmetic means of X and Y series are 6 and 8 respectively.

(c) Obtain the Spearman's rank coefficient of correlation between the variables X and Y from the following pairs of observed values.

X	50	55	65	50	55	60	50	65	70	75
Y	110	110	115	125	140	115	130	120	115	160

OR

Q-6

(7+7)

(a) The following regression equations were obtained from a correlation table:

$$y = 0.516x + 33.73, x = 0.512y + 32.52$$

Find the value of (a) the correlation coefficient,

(b) the mean of x 's and

(c) the mean of y 's.

(b) Find the correlation coefficient and equation of regression lines from the following data:

x	1	2	3	4	5
y	2	5	3	8	7